

THE STORY OF THE ENGINE

THE DRIVING FORCE BEHIND INDUSTRY, ENGINES POWER AN ARRAY OF MACHINES RANGING FROM CARS TO ROCKETS

Engines burn fuel to create hot gases, which expand powerfully to create the mechanical force needed to make things move. As they have developed through the centuries, engines have taken on different forms, from steam engines to rotary engines and gas turbines.

Greek thinkers noticed 2,400 years ago that heat could make things move. In the 1st century CE, they created the aeolipile, in which steam jetted out from a metal sphere to make it spin on a pivot. It was another 1,600 years before the first practical steam engine was built. The breakthrough came in the 1670s, with the discovery of the power of

vacuums. French inventor Denis Papin realized that if steam is trapped in a cylinder, it will shrink dramatically as it condenses to create a partial vacuum powerful enough to move things. In 1698, English inventor Thomas Savery built the first full-scale steam engine using this principle.

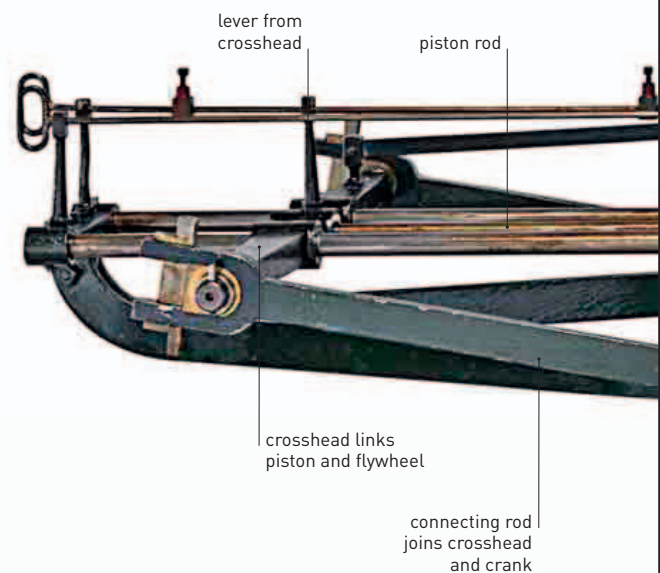
FROM CARS TO MARS

For 150 years, engines all depended on steam. They drove the Industrial Revolution, providing power for everything from machines to ships and locomotives. In the mid-19th century, engineers began to develop internal combustion engines based on the rapid expansion of gases burning inside a cylinder. More compact, these engines used gasoline—a more concentrated source of fuel, which could be drawn in automatically, unlike coal, which had to be manually added. Internal combustion engines were key to the development of automobiles, which transformed mobility in the 20th century. The development of jet and rocket engines helped flying machines achieve previously unimaginable speeds, and eventually, propelled spacecraft to the Moon and beyond.



HYBRID VEHICLES

Heat engines burn a lot of fuel and produce waste gases. Fuel shortage and environmental concerns have led to the development of hybrid vehicles that combine different power sources, such as internal combustion engine and an electric motor, to provide a greener, more cost-effective compromise.



1st century CE

Aeolipile

Alexandrian scholar Hero designs a device in which a sphere is spun by steam jets. A scientific curiosity, it has no apparent practical purpose.

Aeolipile



1712

Newcomen engine

British inventor Thomas Newcomen builds a steam engine that avoids the danger of explosion by boiling water separately, and sending steam at low pressure into a cylinder with a piston.

Newcomen engine



1791

Barber gas turbine

English inventor John Barber patents a gas turbine, intended to propel a "horseless carriage." In this device, fuel is mixed with air and ignited to produce hot gases, which expand and spin a turbine.

1804

Trevithick engine

Low-pressure vacuum-based engines are big and heavy. English engineer Richard Trevithick's develops a compact and powerful high-pressure steam engine.



Steam digester

1679

Papin's steam digester

French inventor Denis Papin invents the steam digester, which traps steam inside a cylinder. This creates a powerful vacuum as the steam cools, condenses, and shrinks.

1698

Savery engine

Thomas Savery builds the first steam engine, to pump water out of mines. It is, however, prone to exploding.

Savery engine



1774

Watt engine

Scottish engineer James Watt produces an improved steam engine, which has a separate condensation chamber, and is more efficient.

Watt engine

